

Project 19: Anomaly Detection Using Autoencoders

Source: [KDD Cup 1999 Dataset](#)

Description

Students will develop an anomaly detection system using autoencoder neural networks and the KDD Cup 1999 dataset. Anomaly detection involves identifying patterns in data that deviate from expected behavior.

The project will begin with an overview of the KDD Cup 1999 dataset, which includes network traffic data with different types of attacks. Students will preprocess the data, including normalization and feature engineering, and implement autoencoder models for anomaly detection.

Training the models involves optimizing hyperparameters, using reconstruction loss functions, and evaluating performance using metrics such as precision, recall, and F1 score. Students will also visualize the reconstruction error distribution and analyze the model's performance qualitatively.

Applications of anomaly detection in real-world scenarios, such as cybersecurity, fraud detection, and system monitoring, will be explored. Students will also discuss the challenges of dealing with imbalanced datasets and noisy signals and propose potential improvements.